Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	11
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1994.91	1519	1026.72	px^2
major_axis_length_px	lacuna shape	85.7455	86.1566	22.5145	px
minor_axis_length_px	lacuna shape	30.4546	26.8577	8.8339	px
aspect_ratio	lacuna shape	2.9668	3.0363	0.8704	unitless
eccentricity	lacuna shape	0.92	0.9442	0.0643	unitless
solidity	lacuna shape	0.8555	0.8762	0.0782	unitless
roots_count	headline	14.0909	13	4.4374	count
ring_length_r30_px	headline	423.091	442	90.8366	px
ring_length_r60_px	reported	989.091	998	188.498	px
owned_length_px	ownership-dependent	2954.87	2421.54	1411.64	px
edge_count	ownership-dependent	138.273	103	69.933	count
mean_edge_length_px	ownership-dependent	21.6005	21.4048	1.8936	px
perimeter_px	normalised	221.572	210.244	51.1926	px
ring_area_r30_px2	normalised	8909.82	8646	1432.67	px^2
ring_area_r60_px2	normalised	22488.8	22056	3166.3	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0478154	0.0490921	0.00947425	px^-1
ring_density_r60	normalised	0.0442888	0.0452485	0.00791038	px^-1
roots_per_100px_perimeter	normalised	6.3641	5.9626	1.4525	per 100 px
ring_attached_length_r30_px	network v2	389.818	399	93.1051	px
ring_attached_length_r60_px	network v2	824.182	828	220.279	px
ring_length_w_r30_px	network v2	479.951	514.316	101.751	px
ring_length_w_r60_px	network v2	1119.82	1128.31	211.225	px
sholl_crossings_r10	network v2	10.1818	10	2.6389	count
sholl_crossings_r20	network v2	12.6364	13	2.6181	count
sholl_crossings_r30	network v2	13.8182	13	3.2193	count
loop_count	loops, report only	13.0909	10	10.2123	count
loop_edge_share	loops, report only	0.3429	0.3495	0.1099	share
ring_width_mean_r30_px	width	6.8505	6.952	0.5632	px

Per field
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measure	value	units
analysed_area_px2	1.02271e+06	px^2
total_skeleton_length_px	41329	px
canalicular_length_density_per_px	0.0404112	px^-1
junction_count	987	count
junction_density_per_px2	0.00096508	px^-2
lacuna_count	14	count

lacunae_per_px2	1.369e-05	px^-2
skeleton_component_count	242	count
mean_component_length_px	170.781	px
median_component_length_px	16.5	px
field_length_density_w_per_px	0.0457258	px^-1
field_density_without_flagged_per_px	0.0401152	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	530	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0404112	px^-1
field_length_density_with_unattached_per_px	0.0409294	px^-1
loop_count	218	count
loop_edge_share	0.3908	share
edge_count	2142	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8295	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	38500	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	510	36.0388	20.5481	1.7539	0.8215	0.8095	-0.2657	13.3863	106.873	True
2	448	43.7126	19.8458	2.2026	0.891	0.6767	1.435	7.0045	363.067	True
3	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
4	1062	83.8665	19.2697	4.3522	0.9732	0.6923	-1.1353	135.124	920.395	False
5	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
6	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
7	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
8	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
9	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
10	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
11	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
12	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
13	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
14	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False

Network per lacuna

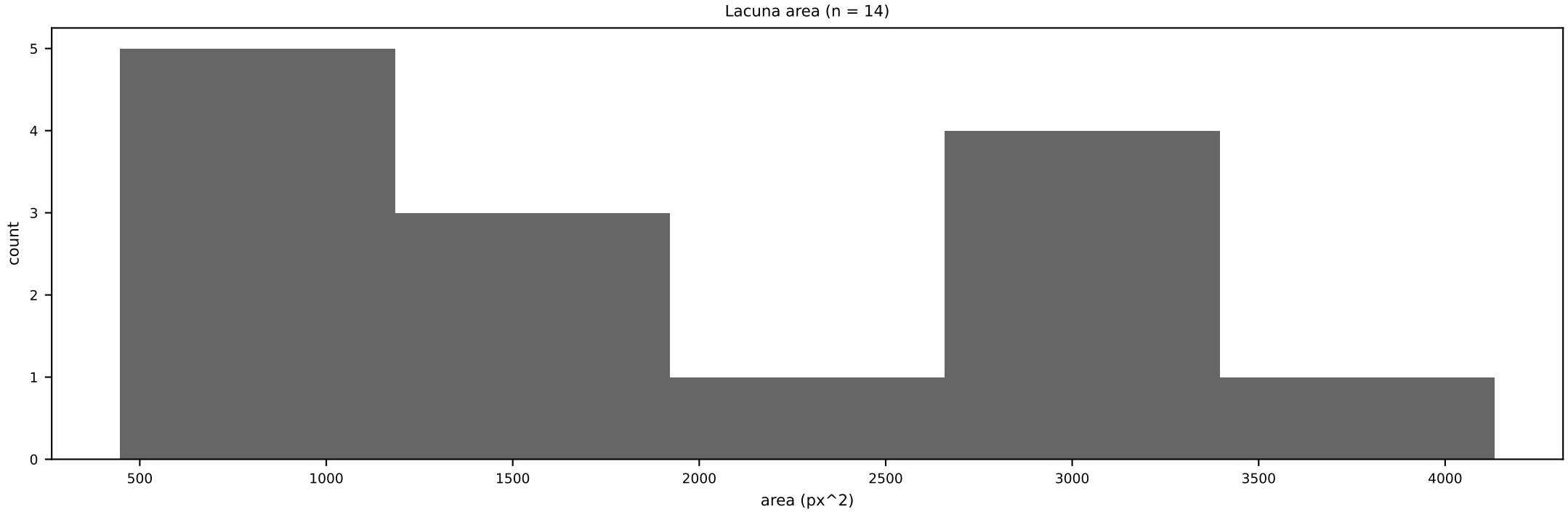
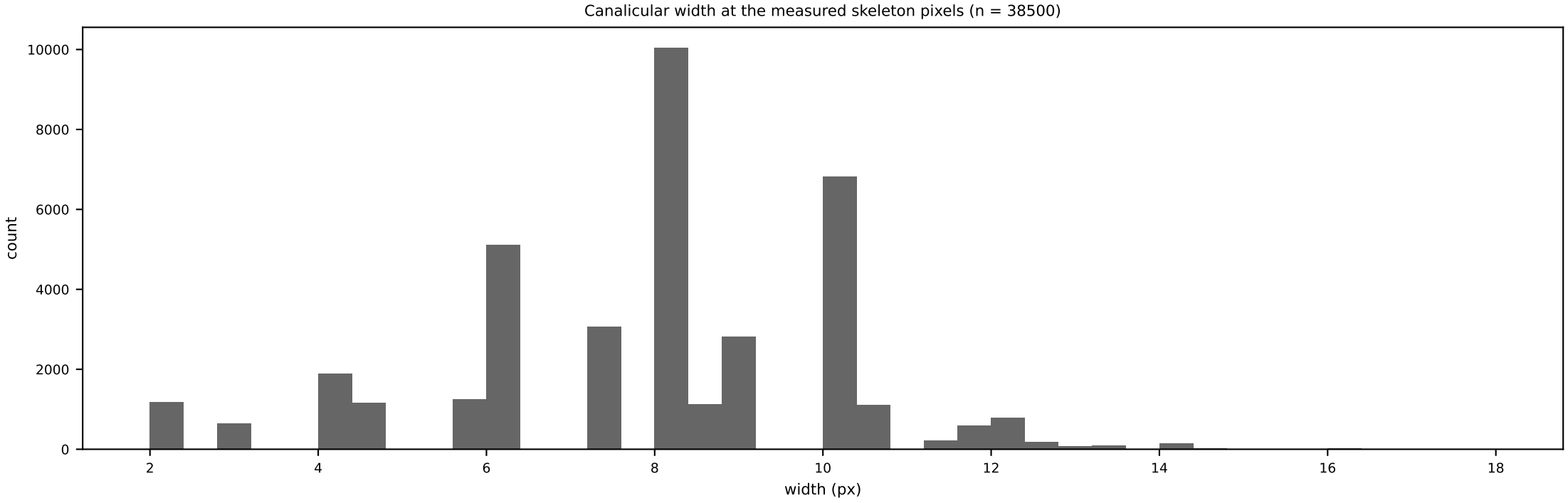
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	160	449	3147.48	160	19.6717	
2	6	77	223	1381.47	78	17.7111	
3	10	400	825	1845.5	95	19.4263	6.0035
4	11	482	1109	5491.25	276	19.8958	6.952
5	20	537	1190	4775.42	216	22.1084	7.3889

6	8	421	1046	4046.57	198	20.4372	7.6867
7	15	465	998	4104.65	193	21.2676	6.8683
8	13	460	980	1999.21	101	19.7941	7.4533
9	13	351	815	1097.88	52	21.1131	7.4606
10	15	546	1365	2421.54	103	23.5101	7.0158
11	13	311	739	2780.9	107	25.9898	6.7399
12	18	442	1011	1505.18	68	22.135	6.0426
13	10	267	790	1391.31	65	21.4048	6.9734
14	22	323	827	2142.01	99	21.6365	6.2309

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	105.154	3701	10161	0.6475	0.597	0.0432316	0.0441886	7.6079
2	132.847	3699	7686	0.6031	0.5656	0.0208164	0.0290138	4.5165
3	219.681	8976	20715	1	0.9689	0.0445633	0.0398262	4.5521
4	210.244	8511	22591	1	1	0.0566326	0.0490903	5.232
5	335.421	12178	29932	1	1	0.0440959	0.0397568	5.9626
6	167.238	7485	20582	1	1	0.0562458	0.0508211	4.7836
7	243.522	9472	22056	1	1	0.0490921	0.0452485	6.1596
8	167.924	7349	17555	1	1	0.0625935	0.0558246	7.7416
9	282.894	8678	19332	0.813	0.7462	0.0404471	0.0421581	4.5954
10	263.48	10172	24552	1	1	0.0536768	0.0555963	5.693
11	174.51	7570	20770	1	1	0.0410832	0.0355802	7.4494
12	210.167	8646	22189	1	1	0.0511219	0.0455631	8.5646
13	185.823	7980	21610	1	1	0.0334586	0.0365571	5.3815
14	259.279	9669	24825	1	1	0.0334057	0.0333132	8.4851

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	145	372	175.468	498.019	5	5	6
2	70	158	87.0122	245.024	2	1	3
3	314	530	444.291	911.34	10	12	15
4	477	1084	554.588	1272.64	13	13	12
5	508	942	597.203	1326.49	10	17	15
6	399	841	493.446	1207.89	11	13	13
7	439	828	524.475	1128.31	12	14	12
8	448	951	520.162	1099.67	11	13	15
9	259	547	400.191	919.21	9	12	11
10	499	1228	611.546	1536.79	15	16	21
11	267	627	351.664	853.394	6	12	10
12	384	825	514.316	1148.35	9	12	17
13	257	643	306.179	908.365	8	9	11
14	296	567	361.593	924.826	7	8	11

682_z23: distributions, pre-validation, px